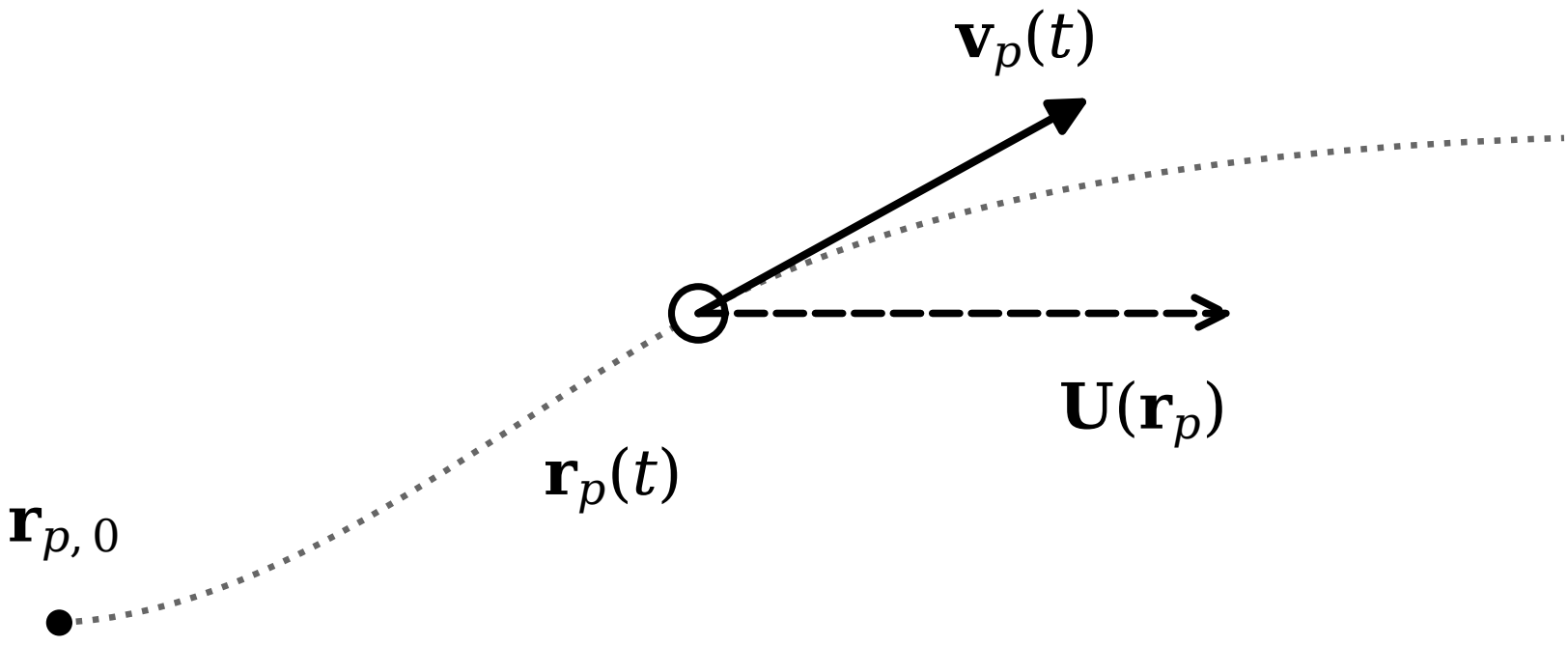
$\mathbf{v}_p(t)$
$\mathbf{U}(\mathbf{r}_p)$
$\mathbf{r}_p(t)$
$\mathbf{r}_{p,\,0}$